Dense Adaptive Grouping Distillation Network for Multimodal Land Cover Classification With Privileged Modality

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***Abstract*— Multimodal land cover classification (MLCC) is a fundamental problem in remote sensing interpretation, which can obtain excellent performance on account of the complementary information between the optical and SAR modalities. However, it is usually impossible to obtain multimodal data at the same time, due to the restriction of imaging conditions. When one of the modalities data is completely missing during the test phase, classical multimodal learning methods might not be able to handle the MLCC task with privileged modality. In this article, we propose an efficient dense adaptive grouping distillation network (DAGDNet), which learns privileged information from available modalities in the train sets and improves the classification performance in the test sets when one of the modality data is scarce. More specifically, to relieve the heterogeneous gaps between different modalities and then transfer the privileged information, we propose an interactive gated-based feature grouping module (IG-FGM), which decomposes multimodal features into modality-shared and modality-specific components to realize the decoupling of multimodal features and grouping distillation. Furthermore, the IG-FGM is inserted into different layers of the "teacher" network to implement progressive blending of multimodalities. Then, to adaptively highlight the importance of hierarchical features distillation and grouping distillation, we propose a multistage adaptive distillation learning (MS-ADL) strategy so that the weights of different distillation losses are required to change continuously along with the training process. Finally, we evaluate the superior performances of our model on representative coregistered optical and SAR datasets.**

***Index Terms*— Deep convolutional neural network (DCNN), feature grouping, knowledge distillation, multimodal land cover classification, privileged information.**

I. Introduction

LAND cover classification under remote sensing (RS) images has received continuous attention, due to its wide applications in the agricultural resource census [1], urban planning [2]–[4], building extraction [5], forest ecological monitoring [6], and so on. Furthermore, with the improvement of Earth observation (EO) technologies, it is easier to obtain different types of RS images from different sources in the same area, e.g., the hyperspectral and optical image consisting of rich spectral and spatial information for land types [7], [8] and the synthetic aperture radar (SAR) data with all-day and all-weather attributes. The complementary information of land properties between different modalities, such as the spectral information of the optical images and the scattering information of the SAR images [9], provides an opportunity to further improve the performances of land cover classification [3], [4], [10]. However, multiple modalities from the same areas are not always simultaneously acquired (as shown in Fig. 1) for some time-critical tasks, such as disaster monitoring [11] and damage assessment [12], due to the restriction of imaging conditions. Also, classical multimodal learning methods inherently have required that all modalities are available during training and testing. These factors constitute the challenges to create fine and timely land cover maps when one of the modalities is absent.

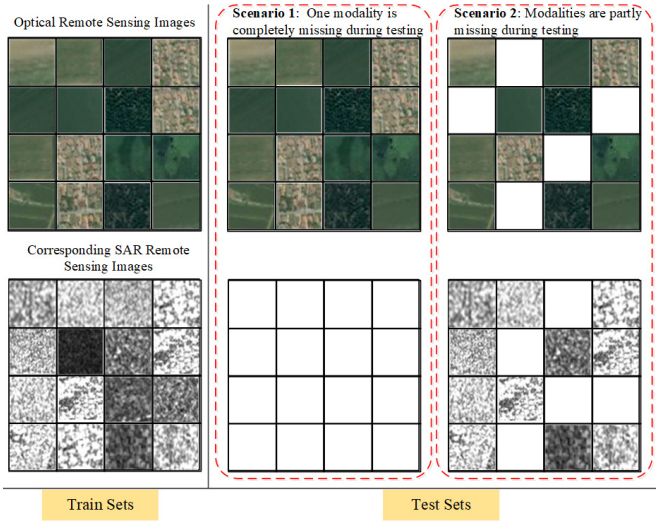

Fig. 1. Two missing situations of test modalities when all training modalities are available. We propose a method to effectively tackle the MLCC when one of the modalities data is completely missing at the test phase.

Learning using privileged information (LUPI) [13] imitated the paradigm of human teaching to guide machine learning based on additional information at training time, which found the best approximates of the unknown decision rule for test sets [14]. However, existing methods either delete extra modality information of train sets and reserve complete modalities to complete the learning of feature extractor and

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classifiers [15], [16] or introduce a series of effective imputation strategies to incomplete modalities based on available modalities [17]. Deleting extra modality is lost the large amounts of potential information, and there is a prerequisite for imputation methods to obtain well results: the modalities data are missing at random (MAR) [17]. The LUPI has been deeply analyzed theoretically and comprehensively applied to computer vision tasks, such as objects recognition with different types of privileged information: ranking with bounding boxes, image tags, and rationales [18]; self-paced learning with poetic descriptions as privileged source [19]; facial point detection with head pose and gender as privileged information [20]; image super-resolution with the ground-truth high-resolution images as privileged information [21]; object localization with the parts, attributes, and segmentation maps as privileged sources [22]; person reidentification [23]; and clustering [24].

Recently, knowledge distillation network models [25], [26] have been proposed to learn the privileged information, which obtains outstanding information over using only one modality. However, early knowledge distillation methods have been mainly used to train compact "student" network based on large "teacher" network using the same modalities data and achieve a tradeoff between the accuracy and the efficiency [27], [28]. Although the privileged knowledge distillation models can realize heterogeneity imitation between different modalities data, when one of the modalities data is missing during the test phase in multimodal tasks [25], heterogeneous land cover classification with missing modality still is a challenging task. Knowledge distillation loss between the "teacher" and "student" networks is used to learn privileged information, which is a common and effective strategy [29], [30]. Traditionally, distillation loss of the last feature layer has been used to conduct knowledge transfer, but it ignored the intermediate layers as distillation hints [30]. After that, hierarchical features between the "teacher" and "student" networks as the hints, which have been significant for efficient knowledge transfer, and the hierarchical privileged information learning methods have achieved excellent performance [31]. Meanwhile, to relieve the settings of hyperparameters and then balance several losses, the adversarial learning paradigm has been used to guide the "student" network to mimic missing modalities [32]. Furthermore, different types of information are used as the knowledge distillation basis that boosted the "student" network to mimic the missing modalities from the "teacher" network, e.g., structured distillation schemes including the pair-wise distillation and the holistic distillation for structured prediction of semantic segmentation [33], intraclass feature variation distillation [34], geometry-aware distillation for action recognition [35], and autism spectrum disorder screening [36].

Knowledge distillation strategies were investigated into multimodal land cover classification (MLCC) with missing modality [37]– [39]. Inspired by literature [40], Kampffmeyer *et al.* [38] proposed the modality hallucination method for urban land cover classification with missing data modalities, which approximated the missing modalities with all available information in training sets, and the experiments proved that the hallucination module can fill in the missing information to a certain extent and improve the classification. Furthermore, to relieve the settings of parameters corresponding to multiple loss functions to balance the total loss, Kampffmeyer *et al.* [38] proposed a universal hallucination network architecture, which adopted a phased learning strategy to guide hallucination network to learn privileged information for efficient MLCC with missing data. Meanwhile, Pande *et al.* [39] explored the deep generative model to dispose modality distillation and then obtain privileged information for MLCC. Although the studies [37]– [39] have achieved promising experimental results, most of them transfer natural image-oriented knowledge distillation methods [26], [31], [33] to achieve MLCC with missing data. Considering that the studies [26], [31], and [33] are mainly used to realize the transfer between homogeneous information, therefore the studies [37]–[39] may not be optimal strategies that are directly used in MLCC with privileged modality, due to the heterogeneous gap between the optical and SAR modalities data. To mitigate semantic divergence between multimodal heterogeneous optical and SAR modalities data, we divide multimodal features into modality-specific and modality-shared components to realize feature grouping knowledge distillation and then use privileged modality to improve the performance of land cover classification.

In this article, a dense adaptive grouping distillation network (DAGDNet) is proposed for land cover classification with privileged modality using heterogeneous optical and SAR images. The interactive gated-based feature grouping module (IG-FGM) groups multimodal features into modality-specific and modality-shared components to realize grouping knowledge distillation, which fairly encourages the distillation of different components to reduce the components' confusion and produce well distillation performance. Meanwhile, the IG-FGM has been embedded into varying layers between the optical and SAR branches on the "teacher" network to progressively explore the interaction of multimodal features. Also, multistage adaptive distillation learning (MS-ADL) consists of component distillation that relieves the semantic gap between heterogeneous features and the hierarchical features distillation that is capable of performing multilayers knowledge distillation. Finally, we indicate the fruitfulness of our method on broad coregistered datasets. Fig. 2 shows the overall architecture of our DAGDNet.

The main contributions are summarized as follows. First, we propose an IG-FGM based on the split-and-aggregation strategy, and its purpose is to explicitly discover modality-shared and modality-specific components to realize grouping knowledge distillation and relieve the semantic gap between heterogeneous features. Furthermore, the IG-FGM is embedded into different layers of the "teacher" network to progressively explore the interaction between multimodal features. Second, we propose an MS-ADL strategy that consists of component grouping distillation that adaptively highlights the contributions of different component distillations for the learning of the "student" network and the hierarchical feature distillation that performs adaptive multilayers knowledge distillation. Finally, we build a DAGDNet framework by

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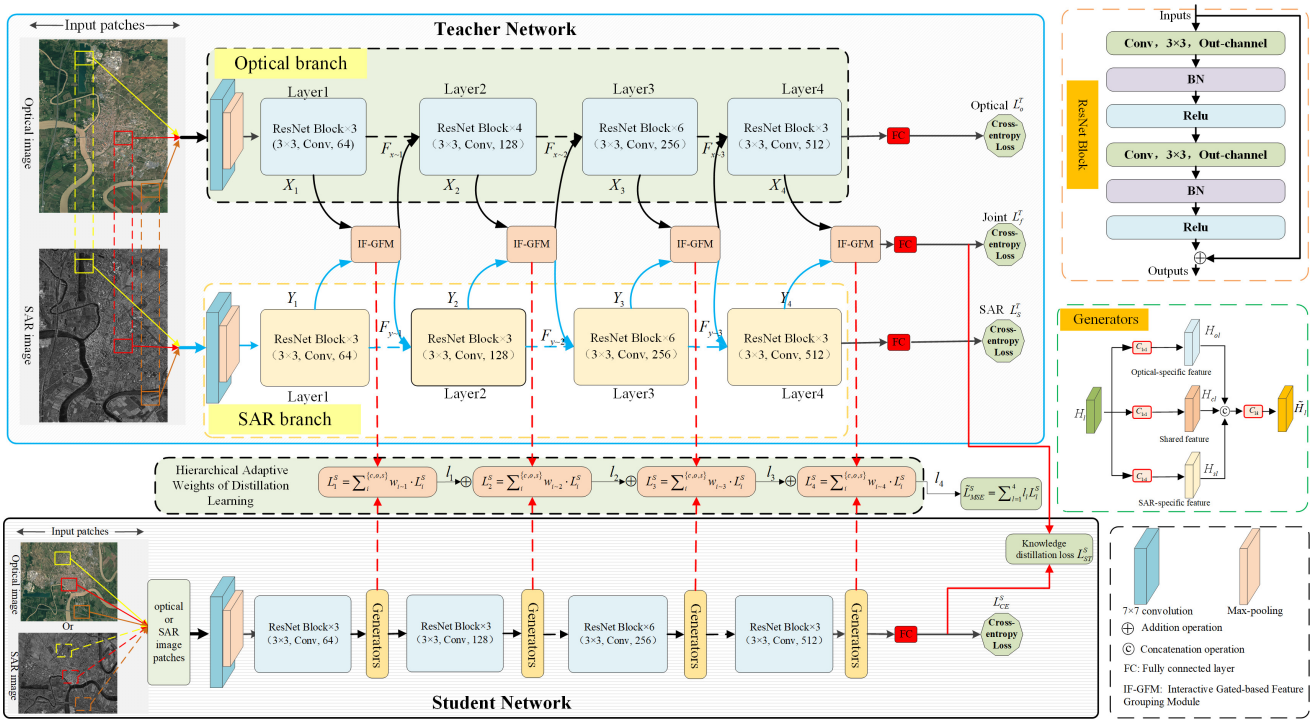

Fig. 2. Overall framework of our proposed DAGDNet for land cover classification with privileged modality (SAR or optical modalities). Note that the DAGDNet includes the IG-FGM and the MS-ADL strategy. The IG-FGM decomposes multimodal features into modality-shared and modality-specific components to realize grouping distillation, and it is also embedded into varying layers of the network for progressive interaction. The MS-ADL emphasizes useful grouping distillation and multilayers distillation.

integrating the IG-FGM and the MS-ADL strategy in an end-to-end manner, which products land cover maps when one of the modalities data is completely missing during the test phase. Furthermore, we prove the fruitfulness of our method on representative coregistered datasets.

II. Related Work

A. Multimodal Land Cover Classification

As two main tools for EO, the optical or SAR images have been widely used in the land cover classification. The traditional unimodal land cover classification methods based on the optical or SAR RS images mainly include three operation steps: 1) data preprocessing which completes radiometric correction, noise reduction of RS images, and so on; 2) handcrafted features extraction, e.g., scale-invariant feature transform (SIFT) and histogram of oriented gradient (HOG) features; 3) classifier design, e.g., support vector machine (SVM) and random forest (RF). Rao *et al.* [41] adopted gray-level cooccurrence matrix (GLCM) to extract textural features from Indian Remote Sensing (IRS)-1D panchromatic data for land cover classification. Traditional unimodal land cover classification methods achieved good results in specific application scenarios, but with the advent of the era of high-resolution RS, the handcrafted features have become more and more powerless for land cover classification, due to the defects of handcrafted features, long design cycle, and weak expression performance. Deep learning is successful technology in computer vision, which integrates features extraction and classifier design in the same frame and achieves better land cover classification performance than traditional unimodal land cover classification methods. Liu *et al.* [3] proposed a statistical convolutional neural network (SCNN) model, which explored the role of statistics information of deep feature in land cover classification with SAR images.

However, the above methods ignored complementary information of the optical and SAR images for the same land types, e.g., the optical images record the spectral information of land objects and the SAR images record the scattering information of land objects. Recently, MLCC is a very active research field, and plenty of methods have been proposed in recent years [42]–[44], which have assigned the semantic labels to whole images. Coregistered multimodal data, such as paired optical-SAR data [45]- [47], paired optical-light detection and ranging (LiDAR) data [48], and triple LiDAR-hyperspectral-optical data [49], which are commonly used multimodal data in the MLCC. The complementary information of multimodal data drives us to extract and fuse relevant multimodal features to enhance the land cover classification performance. Hänsch *et al.* [49] designed a novel node test within the RF framework, which was used to land cover classification using the complementary between the optical images, hyperspectral images, and LiDAR data. Audebert *et al.* [50] investigated the advantages and disadvantages of early and late fusion strategies for urban segmentation using paired optical-LiDAR data. In addition, paired optical-SAR modality-based methods also obtain increasing attention, Li *et al.* [43] explored the second-order fusion strategy instant of linear fusion to extract the interaction between the optical and SAR features, which has obtained

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outgoing MLCC. Steinhausena *et al.* [51] applied the RF-based wrapper approach to select the most relevant radar (Sentinel-1) images for the combination with the optical (Sentinel-2) data, which aimed to generate a land use and land cover map of the Chennai Basin. More recently, attention mechanism plays a more important role in the extraction of complementary information. Li *et al.* [46] applied collaborative attention to realize mutual guidance between the optical and SAR features and obtain the improved MLCC. In this article, we construct an IG-FGM based on a split-and-aggregation strategy, which groups multimodal features into modality-shared and modality-specific components and then progressively explores the interaction between the optical and SAR features from different layers.

B. Knowledge Distillation

Knowledge distillation [29] monitors the learning of the "student" network by migrating helpful knowledge from well-trained "teacher" networks, which have been widely used in semantic segmentation [34], images classification [52], and object detection [40]. The main purpose of early knowledge distillation network models is model compression. Liu *et al*. [33] proposed a structured knowledge distillation network for semantic segmentation, which explored two structured distillation schemes to transfer the structured knowledge to small networks from large networks. The knowledge distillation belongs to the same learning paradigm as LUPI, which has guided the "student" network to learn the privileged information from the "teacher" network. A hallucination branch was designed, which was embedded into the "student" network to learn side information [53]. On that basis, Li *et al*. [52] proposed a dynamic-hierarchical attention distillation network for land cover classification with missing heterogeneous images, which adaptively highlighted different contributions of multilayer attention distillation to promote the hallucination breach. To alleviate the weak robustness issues of existing vision-based action recognition, Liu *et al*. [31] adaptively transferred and distilled the knowledge from multiple wearable sensors to enhance action recognition with vision-sensor modality. Lee *et al*. [53] introduced a distillation framework for efficient image super-resolution, in which the ground-truth high-resolution images were used as the privileged information. To alleviate the time-consuming and labor-intensive of manually labeled privileged information, Yao *et al*. [55] independently learned specific-category classifiers using selected privileged information for each subcategory and then merged them to form a more powerful category classifier. Differently, most of the above knowledge distillation methods used the features from the last softmax layer as the guiding message for privileged information learning. In this article, we introduce an MS-ADL strategy, which not only can automatically encourage the distillation of modality-shared and modality-specific components to reduce the confusion of components and refine the important components for predicting semantic concept but also automatically highlight the distillations of different layers and produces outgoing performances.

C. Feature Grouping

Feature grouping automatically finds highly related feature groups from the data, which has been extensively utilized in the learning of high-dimensional data before the deep learning methods. A naive random grouping is a simple strategy, but it may introduce additional feature grouping bias without prior knowledge. In classical feature grouping methods, the feature grouping techniques based on the fused Lasso families (such as the fused Lasso [56], generalized fused lasso (GFLasso) [57] and discriminative feature grouping (DFG) [58]) as the representation, which assumed the features with identical coefficients from a feature group. More recently, some deep learning methods start to use feature grouping in computer vision tasks [59]- [62]. The interleaved group convolution families solved the redundancy problem in the basic convolution unit of the neural network and increased the calculation speed without compromising performance [59]. Zhao *et al*. [60] proposed an end-to-end grouping recurrent learning (GRL) model that took advantage of the intragroup mutual exclusion and intergroup correlation to improve the pedestrian attribute recognition performance. Zhang *et al*. [61] proposed an unsupervised saliency detection approach, which exploited the grouping and compactness characteristics of the high-level semantic features. Kobayashi *et al*. [62] separated feature components into several groups for encouraging feature learning across all feature components in different losses. We propose an IG-FGM, which groups multimodal features as modality-shared and modality-specific components based on the given grouping weights of each channel feature, and the weights indicate the degree of belonging to the groups.

III. Methodology

A. Framework Overview

We describe a DAGDNet for land cover classification with privileged modality in detail, as shown in Fig. 2. First, we define the "teacher" network architecture that includes two independent optical and SAR branches. It is worth mentioning that multiple IG-FGMs are introduced into different layers of the "teacher" network, and it can not only realize hierarchical interaction between different modalities to improve the performance of the "teacher" network but also divide different modalities features into modality-specific and modality-shared features to complete grouping distillation. Then, to guide the learning of the "student" network, an MS-ADL strategy is proposed for the adaptive weighting of grouping distillation and hierarchical features distillation losses, which explores the different contribution of different components and multilayers features on the distillation process. Finally, the stage learning strategy is used as the training paradigm.

B. Interactive Gated-Based Feature Grouping Module

It is advisable to improve the representation learning capacity of multimodal "teacher" network that can guide the "student" network to learn the privileged information. We introduce a plug-and-play module, namely, an IG-FGM, to facilitate information exchange between the optical and SAR branches in the "teacher" network. Furthermore, existing

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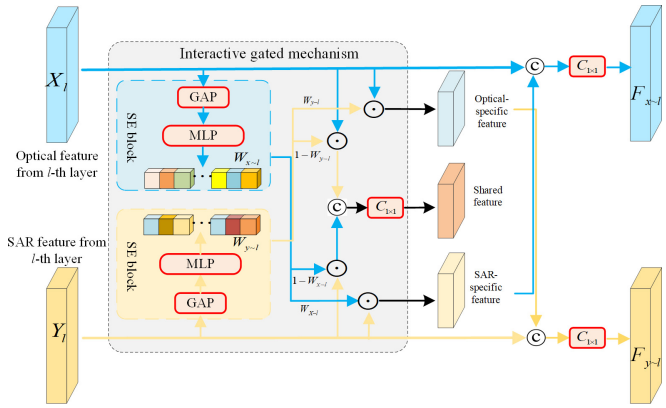

Fig. 3. Structure of the IG-FGM. The MLP denotes Multi-Layer Perception with a hidden layer, the GAP denotes Global Average Pooling, and the SE block denotes the squeeze-and-excitation block. Section III-B describes more details.

knowledge distillation approaches applied to privileged learning that usually conducts privileged knowledge transfer by mimicking privileged modality with the existence of modality in the "student" network, it is not an optimal way that leads to inefficient distillation due to the semantic gap between the heterogeneous features. We insert the proposed IG-FGM module into the "teacher" to learn the modality-shared and modality-specific components based on an interactive gate mechanism. Component features are used as the privileged information that fairly encourages the distillation of different components to reduce the confusion of components. We illustrate the IG-FGM in Fig. 3.

Given the optical-SAR features $X_l, Y_l \in R^{H_l\times W_l\times C_l}$ from the lth layer in the "teacher" network, where H, W, and C represent the height, width, and channel of features, respectively, the IG-FGM is described as follows.

First, to obtain the optical- and SAR-specific features $\tilde{X}_l$ and $\tilde{Y}_l$, inspired by an insight that different channels of feature maps represent different patterns, the optical and SAR features X_l and Y_l are separated based on the interactive gated strategy. Specifically, the optical feature X_l is inputted into the squeeze-and-excitation block and obtains channel selection maps $W_{x\sim l}$ for the SAR feature Y_l, and the SAR-specific feature $\tilde{Y}_l$ is obtained as follows:

$$\begin{cases} W_{x\sim l} = \mathrm{SE}(X_l) \\ \tilde{Y}_l = W_{x\sim l} \odot Y_l \end{cases} \tag{1}$$

where $\mathrm{SE}(\cdot)$ denotes the squeeze-and-excitation block, which consists of a global average pooling layer, two fully connection layers with activation functions, and the Relu and sigmoid functions. $\odot$ denotes the element multiplication operation. Similarly, we can obtain optical-specific feature $\tilde{X}_l$.

Then, to obtain the modality-shared component $\tilde{Z}_l$. We split the modality-shared component $\tilde{Z}_l$ from the optical and SAR features X_l and Y_l based on two ratios $(1 - W_{x\sim l})$ and $(1 - W_{y\sim l})$ as follows:

$$\begin{cases} Z_{x\sim l} = (1 - W_{y\sim l}) \odot X_l \\ Z_{y\sim l} = (1 - W_{x\sim l}) \odot Y_l \\ \tilde{Z}_l = \mathrm{Conv}(Z_{x\sim l} © Z_{y\sim l}) \end{cases} \tag{2}$$

where © denotes the feature concatenation operation and $\mathrm{Conv}(\cdot)$ is the 1×1 convolution operation.

Second, to make the interactions between the optical and SAR branches, we use a split-and-aggregation strategy. The above operations have achieved the separation of multimodal features, and we use the aggregation strategy to interact with the optical and SAR features X_l and Y_l. More specifically, the optical features X_l and the SAR-specific feature $\tilde{Y}_l$ are fused, and the fusion feature $F_{x\sim l}$ is regarded as the input to $(l+1)^{\text{th}}$ layer in the optical branch. Similarly, the fusion feature $F_{y\sim l}$ is obtained and regarded as the input to the $(l+1)^{\text{th}}$ layer in the optical branch

$$\begin{cases} F_{x\sim l} = \mathrm{Conv}(X_l © \tilde{Y}_l) \\ F_{y\sim l} = \mathrm{Conv}(Y_l © \tilde{X}_l). \end{cases} \tag{3}$$

The IG-FGM is inserted into varying layers of the "teacher" in a plug-and-play way, which implements the hierarchical interaction between the optical and SAR modalities to improve the performance of the "teacher" network.

C. Multistage Adaptive Distillation Learning

Distillation loss conducts knowledge transfer, which guides the "student" to imitate the missing or unavailable modality in the "teacher." Traditionally, distillation loss of the last feature layer is used to conduct knowledge transfer, but it ignores the intermediate layers as the distillation hints. For heterogeneous knowledge problems, such as the MLCC using the optical and SAR modalities, modality-specific knowledge is vital for efficient knowledge distillation [65]. To alleviate the modality divergence between the "teacher" and "student" network, we propose an MS-ADL strategy, which consists of feature grouping distillation that can relieve the semantic gap between heterogeneous features, and hierarchical feature distillation that is capable of performing multilayers knowledge distillation to further increase the "student" network.

First, to realize adaptive weighting grouping distillation, given the shared-, optical-, and SAR-specific features $\tilde{Z}_l$, $\tilde{X}_l$, and $\tilde{Y}_l$ from the lth layer in the "teacher" network, respectively, the features H_l from the lth layer in the "student" is used to learn shared-, optical-, and SAR-specific features $H_{c\sim l}$, $H_{o\sim l}$, and $H_{s\sim l}$ based on the generator. We apply the mean square error (MSE) function on feature grouping distillation

$$\begin{cases} L^S_{c\sim l} = \|H_{c\sim l} - \tilde{Z}_l\|_2 \\ L^S_{o\sim l} = \|H_{o\sim l} - \tilde{X}_l\|_2 \\ L^S_{s\sim l} = \|H_{s\sim l} - \tilde{Y}_l\|_2 \end{cases} \tag{4}$$

where $L^S_{c\sim l}$, $L^S_{o\sim l}$, and $L^S_{s\sim l}$ denote the shared-, optical-, and SAR-specific distillation losses in the lth layer.

To highlight varying contributions of different feature grouping distillation, we review the change rates of each grouping distillation loss value over the training process to obtain the dynamic weight average for $L^S_{c\sim l}$, $L^S_{o\sim l}$ and $L^S_{s\sim l}$. Therefore, the weights $w_{c\sim l}$, $w_{o\sim l}$, and $w_{s\sim l}$ are described as follows:

$$w_{i\sim l} = \frac{\exp(\lambda_{i\sim l}(t-1)/T)}{\sum_j \exp(\lambda_{j\sim l}(t-1)/T)}, \quad \lambda_{i\sim l}(t-1) = \frac{L^S_{i\sim l}(t-1)}{L^S_{i\sim l}(t-2)} \tag{5}$$

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where $i, j \in \{c, o, s\}$, the $\exp(\cdot)$ denotes the exponential function, t is the training epoch, and T represents a temperature that controls the softness of component weights.

We employ adaptive weighting distillation loss terms L_l^S at the lth layer, including the shared-, optical- and SAR-specific distillation losses $L_{c\sim l}^S$, $L_{o\sim l}^S$, and $L_{s\sim l}^S$

$$L_l^S = \sum_i^{\{c,o,s\}} w_{i\sim l} \cdot L_{i\sim l}^S. \tag{6}$$

Second, to realize adaptive weighting of multilayers distillation, we design different layers of adaptive distillation loss terms, which is important for efficient knowledge transfer. To improve the generalization performance of the "student" network, we review the layers' distillation loss values of verification sets as the weights for the corresponding layers' distillation loss in training. Therefore, the adaptive weighting hierarchical distillation is described as follows:

$$\tilde{L}_{\text{MSE}}^S = \sum_l l_l \cdot L_l^S \tag{7}$$

where $\tilde{L}^S$ denotes the adaptive weighting hierarchical distillation loss and l_l denotes the weight value from the verification loss.

D. DAGDNet for MLCC With Privileged Modality

We integrate the IG-FGM and the MS-ADL into an end-to-end knowledge distillation framework, namely, DAGDNet, to address land cover classification with privileged modality. In the DAGDNet, paired optical-SAR images are used as the inputs of the "teacher" network, and one of the modalities data is input into the "student" network.

1) Training of the "Teacher" Network: We build a "teacher" network that consists of independent optical and SAR branches with ResNet34 as the backbone. As shown in Fig. 2, we simultaneously feed paired optical-SAR modalities into the corresponding branches for model training. The IG-FGM is embedded into selected layers of the "teacher" network to realize different modalities' interactions. The "teacher" network is trained simultaneously using the optical and SAR branches losses L_o^T and L_s^T as well as joint cross-entropy loss L_f^T of heterogeneous fusion features. The total loss L^T for the "teacher" networks is defined as follows:

$$\begin{aligned} L^T &= L_o^T + L_s^T + L_f^T \\ &= \text{CE}\left(F_o^T, P\right) + \text{CE}\left(F_s^T, P\right) + \text{CE}\left(F_f^T, P\right) \end{aligned} \tag{8}$$

where $\text{CE}(\cdot)$ denotes the cross-entropy function, P is the true labels, and F_o^T, F_s^T, and F_f^T denote the predicted labels of the optical- and SAR-branch and heterogeneous fusion features, respectively.

2) Training of the "Student" Network: First, we freeze the parameters of the "teacher" networks. Then, we propose an MS-ADL, which not only relieves the semantic gap between heterogeneous features but also is capable of performing multilayers knowledge distillation, further increasing the accuracy of the "student" network.

In addition, we build a knowledge distillation loss at the last fully connected layers between the "teacher" and "student" networks

$$L_{\text{ST}}^S = \text{KL}\left(\frac{F_f^T}{\tilde{T}}, \frac{F^S}{\tilde{T}}\right) \tag{9}$$

where $\text{KL}(\cdot)$ denotes the Kullback–Leibler divergence, F^S is the predicted label of the "student network," and $\tilde{T}$ denotes the temperature to control the distribution of the probability, which is set to 2.

To train the "student" network, we use the cross-entropy loss $L_{\text{CE}}^S = \text{CE}(F^S, P)$ and knowledge distillation loss $\tilde{L}^S$ as well as L_{ST}^S. The total loss L^S for the "student" network is defined as follows:

$$L^S = \alpha L_{\text{CE}}^S + \beta \tilde{L}_{\text{MSE}}^S + \gamma L_{\text{ST}}^S \tag{10}$$

where α, β, and γ are the parameters controlling the importance of the cross-entropy loss $L_{\text{CE}}^S = \text{CE}(F^S, P)$ and knowledge distillation loss $\tilde{L}^S$ as well as L_{ST}^S, respectively.

Algorithm 1 Training/Test Processing of the DAGDNet Method

Input: The coregistered optical and SAR training patches $\{X, Y\}$, and optical or SAR test patches $\tilde{X}$ or $\tilde{Y}$, the maximum iterations I.

Output: MLCC with privileged modality.

1. The training processing of the DAGDNet method:

1.1: Training the "Teacher" network;

1.1.1: Initialize the "Teacher" network weight W_T, learning rate ξ;

1.1.2: **if** $i \leq I$ **then**

1.1.3 Update the network weight W_T based on (8)

1.1.4: end if

1.2: Training the "Student" network;

1.2.1: Freeze the parameters of the "Teacher" networks, and initialize the "Student" network weight W_S, learning rate ξ;

1.2.2: if $i \leq I$ **then**

1.2.3 Update the network weight W_S based on (10)

1.2.4: **end if**

2. The test processing of the DAGDNet method:

2.1: Input the optical or SAR test patches $\tilde{X}$ or $\tilde{Y}$ into trained "Student" network of Fig. 2;

2.2: Output MLCC results with privileged modality.

IV. Experiments and Discussion

A. Experimental Setups

1) Datasets: We estimate the effectiveness of the proposed DAGDNet on the Libourne, Bergerac, and PoDelta datasets, as shown in Fig. 4. The Libourne comes from the region (distributed between the (44.53 °N, 0.10 °E) upper left coordinates and (44.57 °N, 0.16 °E) lower right coordinates) of Libourne city, Bordeaux, France, which includes coupled GoogleEarth and TerraSAR with horizontal transmit-horizontal receive (HH) polarisation imaging mode data acquired on June 28 and September 22, 2009, respectively. Eventually, both GoogleEarth and TerraSAR images have been coregistered and obtained a final size

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TABLE I

LAND TYPES DISTRIBUTION ON THE LIBOURNE, BERGERAC, AND PODELTA DATASETS, AND LAND COVER TYPES IN BRACKETS ARE THE PODELTA DATASET

#	Land Types	Libourne		Bergerac		PoDelta	
		Number	Percentage	Number	Percentage	Number	Percentage
1	Resident area	3999047	17.97%	359938	18.81%	213868	2.99%
2	Lake (Water)	355261	53.58%	210512	1.10%	826053	11.55%
3	Farmland	11890375	53.41%	7081777	37.01%	497418	6.95%
4	Tree	2949197	13.25%	3525715	18.42%	3834510	53.58%
5	Road (Background)	723753	3.25%	620455	3.24%	1784183	24.93%
6	Industrial land	914262	4.11%	674799	3.53%	–	–
7	River	1426033	6.41%	1845398	9.64%	–	–
8	Bare-soil	–	–	1578410	8.35%	–	–

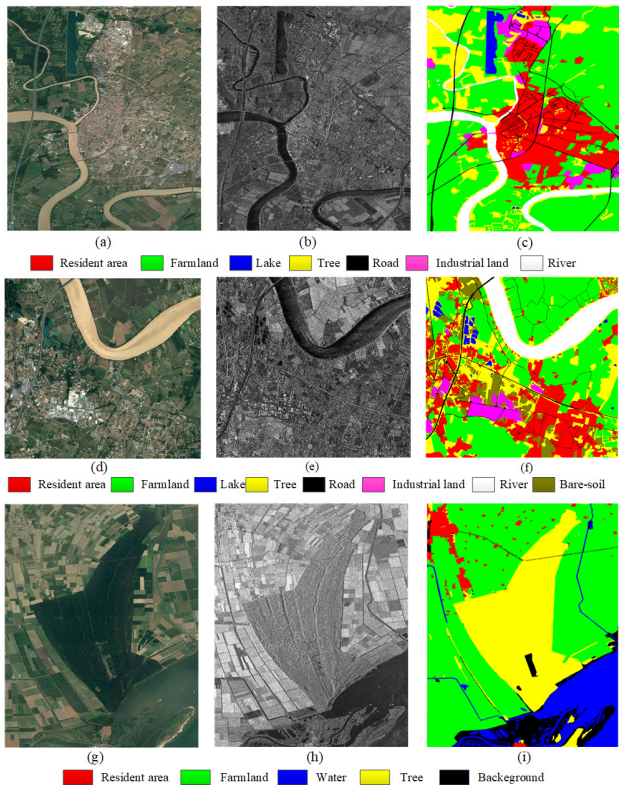

Fig. 4. Visualization of the Libourne, Bergerac, and PoDelta datasets. (a), (d), and (g) Optical images. (b), (e), and (h) SAR images. (c), (f), and (i) Ground truths.

of 6384 × 3542 with 3.0 m × 3.0 m. The ground truth of the Libourne is obtained based on Open Street Maps [64] and visual interpretation. Seven land types are contained in the Libourne: resident area, river, road, farmland, tree, industrial area, and lake.

The Bergerac comes from the region (distributed between the (44.54 °N, 0.24 °E) upper left coordinates and (44.57 °N, 0.28 °E) lower right coordinates) of Bergerac city, France, which includes coupled GoogleEarth and TerraSAR with HH-polar imaging mode data acquired on September 29 and 30, 2016, respectively. Eventually, both GoogleEarth and TerraSAR images have been coregistered and obtained a final size of 4577 × 4181 with 3.0 m × 3.0 m spatial resolution. The ground truth of the Bergerac is obtained based on Open Street Maps [64] and visual interpretation. Eight land types are contained in the Bergerac: resident area, farmland, lake, tree, road, industrial area, river, and bare soil.

The PoDelta comes from a region of PoDelta, Italy, which includes coupled GoogleEarth and COSMO-SkyMed with HH-polar imaging mode data. Both GoogleEarth and COSMO-SkyMed images have been coregistered and obtained a final size of 3666 × 1752 with 3.0 m × 3.0 m spatial resolution. We relabel the ground truth of the PoDelta provided by the literature [43], [46]. Five land types are contained in the PoDelta: resident area, water, farmland, tree, and background.

2) Implementation Details: We take ResNet34 as our backbone network of the DAGDNet. All the experiments are conducted with two NVIDIA RTX 2080Ti graphic processing unit (GPU) using PyTorch, an Adam optimizer with a base learning rate of $\xi = 1e{-}3$, a weight decay $\beta = 4e{-}8$, and a momentum $\eta = (0.90, 0.99)$ is used. Also, we use an exponentially decaying learning rate with a decay rate of 0.95 in later epochs. After completing the training of the "teacher," we freeze trained model parameters of the "teacher" to train the "student" network. Also, the batch size is 128, and the epoch is 100. Considering that the optical and SAR images of both Libourne and Bergerac datasets are collected from the same sensors and have similar land types, and their image patch sizes are all set to 33 × 33, and for the PoDelta dataset, the patch size is to 25 × 25. The numbers of training and validation samples in the Libourne, Bergerac, and PoDelta datasets for each land type are (40 000, 4000), (40 000, 4000), and (25 000, 2500), respectively, and the remainders are used to test sets. The pixel accuracy (PA), mean pixel accuracy (MPA), mean intersection over union (MIoU) and frequency weighted intersection over union (FMIoU) are used as the performance metrics.

B. Hyperparameter Configuration

1) Input Setting of IG-FGM: In our proposed DAGDNet, the IG-FGM aims at implementing progressive interaction between the optical and SAR features to promote land cover classification in the "teacher" network. To determine which layer optical and SAR features should interact, we experiment on the Libourne, Bergerac, and PoDelta datasets and record results in Table II. The optical and SAR branches contain four levels of residual layers. Row#1–row#12 show that an IG-FGM is embedded into different single-level of the optical and SAR branches, respectively, which is used

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TABLE II

CLASSIFICATION PRECISION FOR OUR METHOD WITH THE DIFFERENT AMOUNT OF THE IG-FGM IS INSERTED INTO THE TEACHER

#	Datasets	Number of IG-FGM				Metrice(%)	
		Layer1	Layer2	Layer3	Layer4	PA	MIoU
1	Libourne	✓				88.23	78.39
2	Bergerac	✓				90.01	89.03
3	PoDelta	✓				94.98	86.23
4	Libourne		✓			88.67	78.84
5	Bergerac		✓			90.52	89.21
6	PoDelta		✓			95.02	86.71
7	Libourne			✓		88.71	78.93
8	Bergerac			✓		90.61	89.54
9	PoDelta			✓		95.11	86.44
10	Libourne				✓	89.14	79.32
11	Bergerac				✓	90.92	89.95
12	PoDelta				✓	95.28	86.86
12	Libourne	✓	✓			88.93	79.05
13	Bergerac	✓	✓			90.89	89.71
14	PoDelta	✓	✓			95.25	86.79
15	Libourne	✓	✓	✓		89.61	81.01
16	Bergerac	✓	✓	✓		91.03	89.98
17	PoDelta	✓	✓	✓		96.01	87.22
18	Libourne	✓	✓	✓	✓	**90.89**	**81.35**
19	Bergerac	✓	✓	✓	✓	**91.52**	**90.33**
20	PoDelta	✓	✓	✓	✓	**96.65**	**88.82**

to fuse the single-level optical and SAR features. It is seen that the land cover classification performance has a higher improvement, as the IG-FGM is embedded in a deeper layer, mainly because the optical and SAR features from layer4 have a smaller heterogeneous gap, which is conducive to the exchange of information from heterogeneous sources. For multiple IG-FGMs, they are added from low to high layers in a cascaded manner. As shown in Table II, IG-FGM leads to the mutual flow of information between the optical and SAR branches, which has a positive affection on the interaction between the optical and SAR features. Also, by comparing row#1–row#12 and row#18–row#20, it cannot achieve the same performance when IG-FGM was only inserted into the last layer. As the number of IG-FGM is increasing in the "teacher" network, it is seen that the "student" network has more improvement.

2) Hyperparameters α, β, and γ Setting: To study the effect of three hyperparameters α, β, and γ in (10), we conduct the analysis of the parameters on the Libourne, Bergerac, and PaDelta datasets, respectively. The ranges of parameters α and β are $\{0.1, 1\}$, and the range of parameters γ is $\{0.01, 0.1, 1\}$. Table III shows the performance of our DAGDNet by setting different values to the parameters α, β, and γ. Moreover, we transform the parameter of Table III into a 4-D scatter diagram to further understand the choice of the optimal parameter, as shown in Fig. 5. From Table III and Fig. 5, it is seen that the optimal hyperparameters are $\{1, 0.1, 0.01\}$ in the Libourne dataset, while for Bergerac dataset, the optimal hyperparameters are set to $\{1, 1, 0.1\}$, and, for PaDelta dataset, the optimal hyperparameters are $\{1, 1, 0.01\}$. For parameter α, its optimal values are 1 for all three datasets, which are more than other hyperparameters β and γ. This is because the hard-target loss contributes to the learning of the "student" network, which should be deserved more attention, while the soft-target and the MS-ADL loss only conduct knowledge distillation, which plays the role of a supporter. For parameter β, its optimal values are more than the hyperparameter γ for all datasets, mainly because the soft-target loss only executes knowledge distillation at the last fully connected layers, while the MS-ADL loss contributes to the knowledge transfer of all layers.

TABLE III

PARAMETERS SENSITIVITY ANALYSIS OF α, β, AND γ ON THE LIBOURNE, BERGERAC, AND PODELTA DATASETS

Hyperparameters			Datasets					
α	β	γ	Libourne		Bergerac		PoDelta	
			PA	MIoU	PA	MIoU	PA	MIoU
1	1	1	88.45	79.85	91.04	89.98	94.14	86.98
1	0.1	1	88.80	80.61	91.32	90.12	94.05	86.82
0.1	1	1	89.67	80.86	90.28	89.07	93.95	86.56
0.1	0.1	1	89.78	80.76	90.78	89.74	93.65	86.41
1	1	0.1	90.12	81.17	**91.52**	**90.33**	96.05	87.49
1	0.1	0.1	90.09	81.15	91.42	90.01	96.21	87.53
0.1	1	0.1	89.99	80.32	90.34	89.22	95.78	87.09
0.1	0.1	0.1	90.32	80.67	89.99	89.02	95.62	86.91
1	1	0.01	90.01	80.91	90.98	89.97	**96.65**	**88.82**
1	0.1	0.01	**90.89**	**81.35**	91.13	90.03	96.25	87.75
0.1	0.1	0.01	90.03	81.02	89.03	88.98	95.85	87.05
0.1	1	0.01	90.06	81.14	89.43	89.03	95.35	86.88

3) Temperature Parameters T Setting: The temperature T in (5) is used to control the softness of feature component weights $w_{i\sim l}\,(i \in \{c, o, s\})$, and when $w_{i\sim l}$ is larger, the corresponding $L^S_{i\sim l}$ in (6) is more important, which means that the corresponding feature component (modality-shared component $\tilde{Z}_l$ optical-specific features $\tilde{X}_l$ or SAR-specific features $\tilde{Y}_l$) have more important to guide the learning of the "Student" network in features grouping distillation. Fig. 6 shows the performance with different temperatures T on the Libourne, Bergerac, and PaDelta datasets, respectively. As shown in Fig. 6, the accuracy curve rises with the increase of temperature T, and after reaching a certain peak, it decreases with the increase of temperature T, mainly because with temperature T increase, $w_{i\sim l}$ of different feature component ($\tilde{Z}_l$, $\tilde{X}_l$, and $\tilde{Y}_l$) will tend to flatten, which can make more privileged information from different feature components to guide the learning of the "Student" network. However, too large temperature T will cause feature component weights $w_{i\sim l}$ to converge, which makes the MS-ADL unable to adaptively highlight the contributions of different feature components during distillation learning. Also, it is seen that the classification accuracies are optimal when the temperature T is on the Libourne, Bergerac, and PaDelta datasets.

C. Comparison With State-of-the-Art Methods

We compare the classification performances of our proposed DAGDNet using heterogeneous optical and SAR images with three categories of methods: 1) state-of-the-art multimodal knowledge distillation methods [30], [38], [52] with different backbones, whose classification results are presented in rows #10–#17 of Table IV; 2) unimodal land cover classification methods [3], [66], whose classification results are presented in

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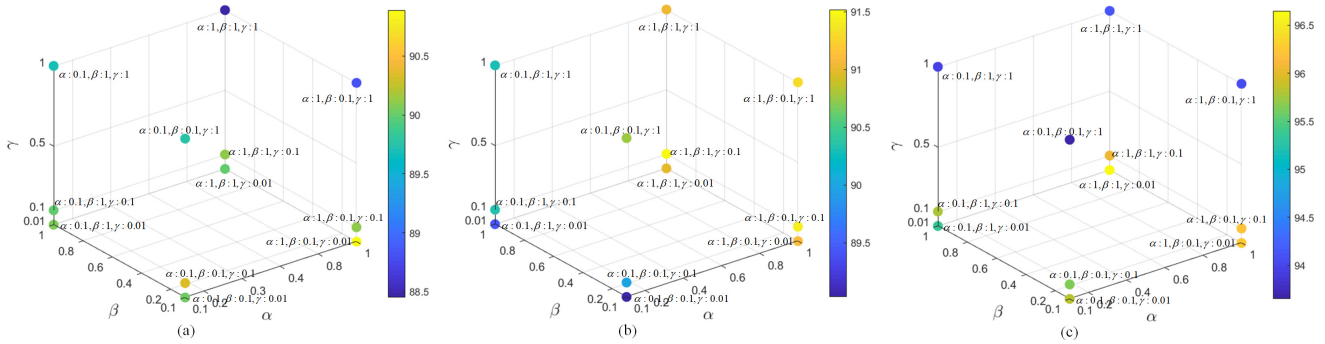

Fig. 5. 4-D scatter diagram for parameter sensitivity analysis of α, β, and γ. (a) Libourne, (b) Bergerac, and (c) PoDelta datasets.

TABLE IV

PA, MPA, MIoU, FWIoU, AND KAPPA OF DIFFERENT METHODS ON THE LIBOURNE, BERGERAC, AND PADELTA DATASETS, RESPECTIVELY. BOLD DENOTES THE BEST RESULTS, AND BLUE DENOTES THE SUBOPTIMAL RESULTS

#	Methods	Backbone	Test Modalities	Libourne					Bergerac					PoDelta				
				PA	MPA	MIoU	FWIoU	Kappa	PA	MPA	MIoU	FWIoU	Kappa	PA	MPA	MIoU	FWIoU	Kappa
1	SCNN [3]	–	Opt.	77.07	81.35	63.01	65.99	67.65	74.40	80.67	67.82	62.62	68.22	84.75	85.52	64.15	77.23	77.38
2	Suoyan *et.al* [66]	–	Opt.	80.70	84.99	71.24	69.94	69.53	80.93	85.84	75.50	68.99	74.85	89.08	87.51	70.26	82.17	82.09
3	2D-CNN [10]	–	Opt.-SAR	81.17	86.09	68.03	68.11	74.28	81.70	87.13	77.88	69.77	78.32	92.54	89.21	82.99	87.01	86.89
4	MDL-RS [45]	–	Opt.-SAR	83.61	89.50	71.16	73.86	76.43	80.93	86.57	77.62	68.96	77.08	92.05	91.64	76.69	86.57	85.75
5	MultiCNN [63]	–	Opt.-SAR	85.01	90.25	72.24	75.19	78.26	85.54	90.58	83.64	75.43	81.61	94.84	94.62	81.77	91.16	88.32
6	MBFNet [43]	AlexNet*	Opt.-SAR	88.31	92.79	77.45	78.92	81.12	88.39	92.43	88.92	79.23	85.76	96.18	96.53	84.12	92.03	91.98
7	MBFNet [43]	ResNet34	Opt.-SAR	89.41	94.24	79.48	81.11	84.50	91.41	93.85	89.84	81.09	89.01	96.77	97.01	87.79	94.15	94.96
8	CHGFNet [46]	AlexNet*	Opt.-SAR	89.31	92.79	77.45	80.89	82.86	89.56	93.41	88.92	79.53	86.76	**97.86**	97.11	88.59	94.54	92.09
9	CHGFNet [46]	ResNet34	Opt.-SAR	**91.04**	**94.89**	**81.48**	81.98	**86.04**	**91.65**	**94.81**	**90.84**	**82.59**	**90.02**	97.25	**97.57**	**89.81**	**94.87**	**95.68**
10	Garcia *et. al* [30]	AlexNet*	Opt.	85.98	89.68	74.03	76.70	80.01	88.66	91.28	87.26	79.91	86.18	94.85	94.74	82.99	91.07	87.21
11	Kampffmeyer *et.al* [38]	AlexNet*	Opt.	85.05	88.26	72.36	75.69	79.05	87.15	90.32	84.80	78.12	83.61	93.58	93.79	81.75	90.02	85.67
12	DH-ADNet [52]	AlexNet*	Opt.	86.95	90.34	75.08	77.92	81.05	89.23	92.01	88.19	80.79	85.87	95.22	95.48	83.47	91.56	84.98
13	DH-ADNet [52]	ResNet34	Opt.	88.24	92.24	79.48	80.10	83.62	90.08	92.99	88.95	81.56	87.34	95.78	95.56	85.78	92.02	93.37
14	Garcia *et. al* [30]	AlexNet*	SAR	73.56	75.58	59.87	60.42	64.01	71.54	80.62	65.66	57.16	65.67	85.96	88.84	71.55	76.77	76.74
15	Kampffmeyer *et.al* [38]	AlexNet*	SAR	72.24	74.57	57.96	57.56	63.25	70.92	80.32	64.89	56.56	64.45	84.31	87.54	67.85	74.29	75.41
16	DH-ADNet [52]	AlexNet*	SAR	74.12	76.25	60.58	61.49	63.79	72.34	81.01	66.67	58.01	64.04	86.99	89.79	72.04	77.98	79.02
17	DH-ADNet [52]	ResNet34	SAR	74.76	76.98	61.13	62.02	65.03	73.81	82.22	67.48	58.98	67.24	87.24	90.01	72.36	78.99	81.20
18	DAGDNet (our)	AlexNet*	Opt.	88.43	91.56	76.81	82.89	82.21	89.87	93.56	88.78	79.04	86.21	95.98	96.48	85.47	92.76	93.45
19	DAGDNet (our)	ResNet34	Opt.	90.89	94.33	81.35	**82.54**	85.45	91.52	94.56	90.33	82.36	89.84	96.95	97.06	88.82	94.09	95.42
20	DAGDNet (our)	AlexNet*	SAR	74.71	76.76	61.12	62.56	64.45	73.09	81.71	67.15	59.02	65.38	87.79	90.08	72.53	79.08	81.01
21	DAGDNet (our)	ResNet34	SAR	75.83	78.54	62.83	63.70	66.24	75.28	84.01	68.54	60.57	69.22	89.03	91.60	74.17	80.79	83.38

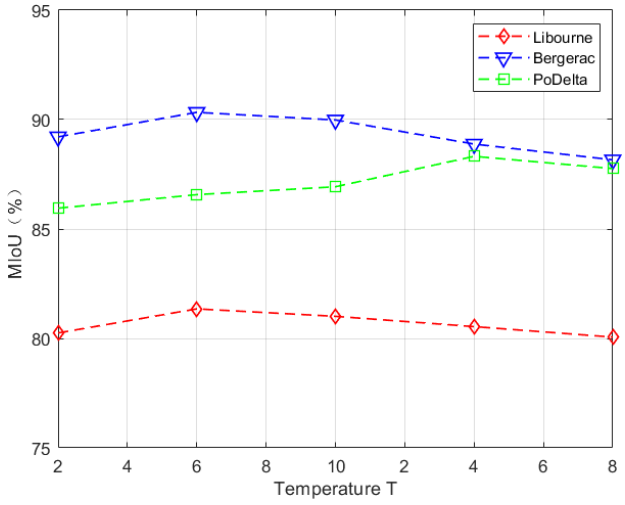

Fig. 6. Classification performance of the proposed DAGDNet with different temperatures T.

rows #1 and #2; 3) MLCC methods [10], [43], [45], [63], [46] with different backbones, whose classification results are presented in rows #3–#9. We compute five widely used criteria: the pixel accuracy (PA), mean pixel accuracy (MPA), mean intersection over union (MIoU), frequency weighted intersection over union (FWIoU), and Kappa indictor. Also, AlexNet* denotes a modified version of original AlexNet classifier, whose structure can refer to the literature [52].

Table IV lists the quantitative classification results of different methods on the Libourne, Bergerac, and PaDelta datasets, respectively. Generally, there is an obvious improvement in four main indices when jointly using both optical and SAR modalities, compared to unimodal land cover classification methods using optical and SAR independently. Moreover, different backbones are adopted, such as AlexNet and ResNet34, and it is seen that a deeper backbone encoder can extract more discriminative features to facilitate classification, i.e., row#6 versus row#7. Meanwhile, we verify the effectiveness of our method at two different missing situations of test modalities when all training modalities can be available, namely, missing SAR modality and missing optical modality. Overall, the classification accuracy of our method with missing SAR modality is higher than that with missing optical modality. The main reason is that the optical data have rich land information and better interpretability that make it become

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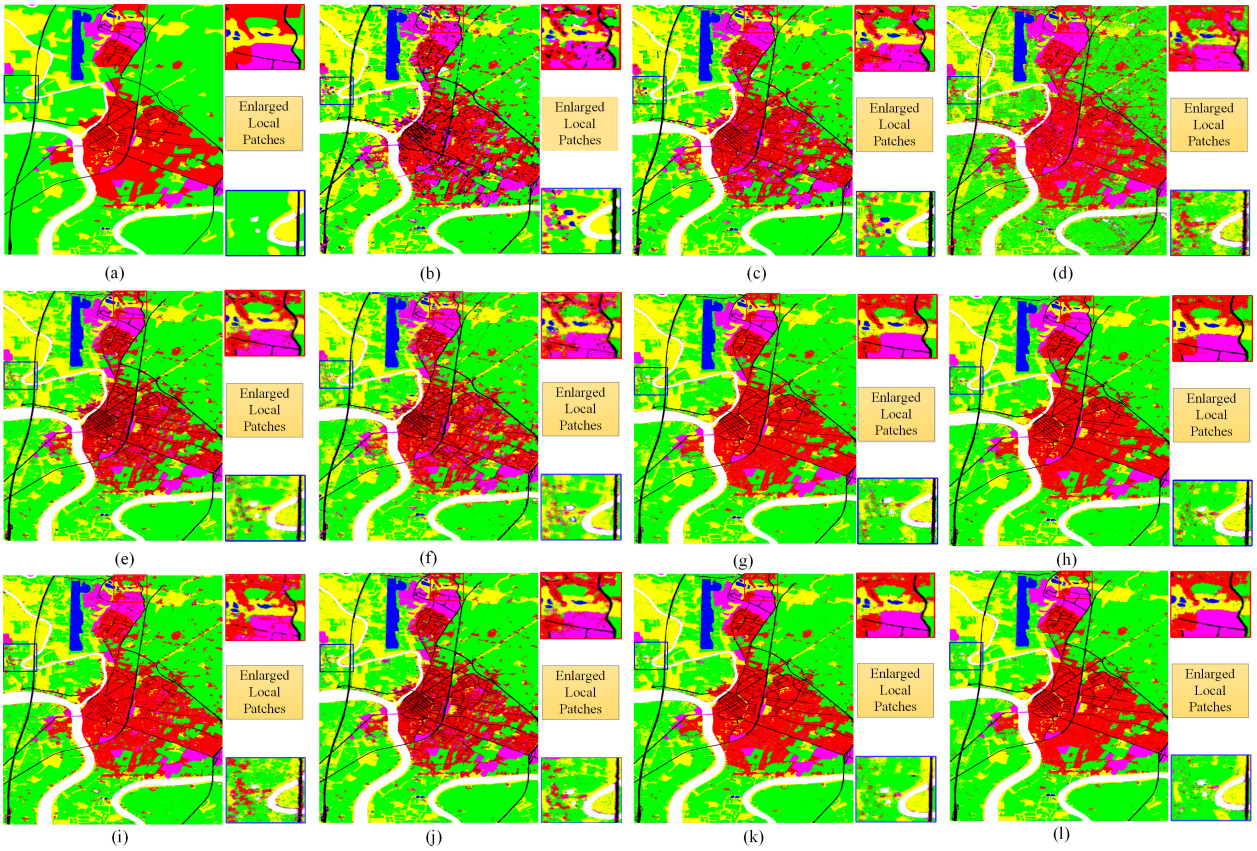

Fig. 7. Classification maps of different methods with missing SAR modality on Libourne. (a) Ground truth. (b) SCNN [3]. (c) Suoyan *et.al* [66]. (d) 2-D-CNN [10]. (e) MDL-RS [45]. (f) MultiCNN [63]. (g) MBFNet [43] (ResNet34). (h) CHGFNet [46] (ResNet34). (i) Garcia *et. al* [30]. (j) Kampffmeyer *et.al* [38]. (k) DH-ADNet [52] (ResNet34). (l) DAGDNet (ResNet34).

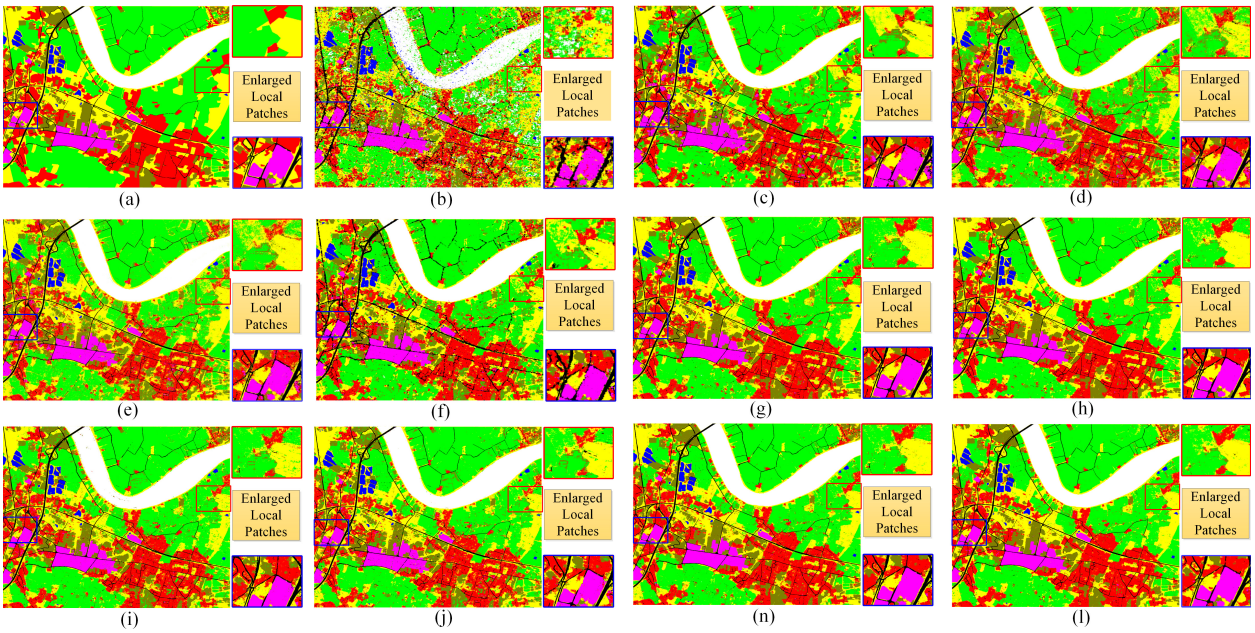

Fig. 8. Classification maps of different methods with missing SAR modality on Bergerac. (a) Ground truth. (b) SCNN [3]. (c) Suoyan*et.al* [66]. (d) 2-D-CNN [10] (e) MDL-RS [45]. (f) MultiCNN [63]. (g) MBFNet [43] (ResNet34). (h) CHGFNet [46] (ResNet34). (i) Garcia *et. al* [30]. (j) Kampffmeyer *et.al* [38]. (k) DH-ADNet [52] (ResNet34). (l) DAGDNet (ResNet34).

the dominant data in land cover classification, as shown in Fig. 11, and the Opt.-branch obtains performance far exceeding the SAR-branch, in which the SAR-branch and Opt.-branch are uinmodal land cover classification methods, which come from the teacher in Fig. 2. In our DAGDNet method, the privileged information that the student learns

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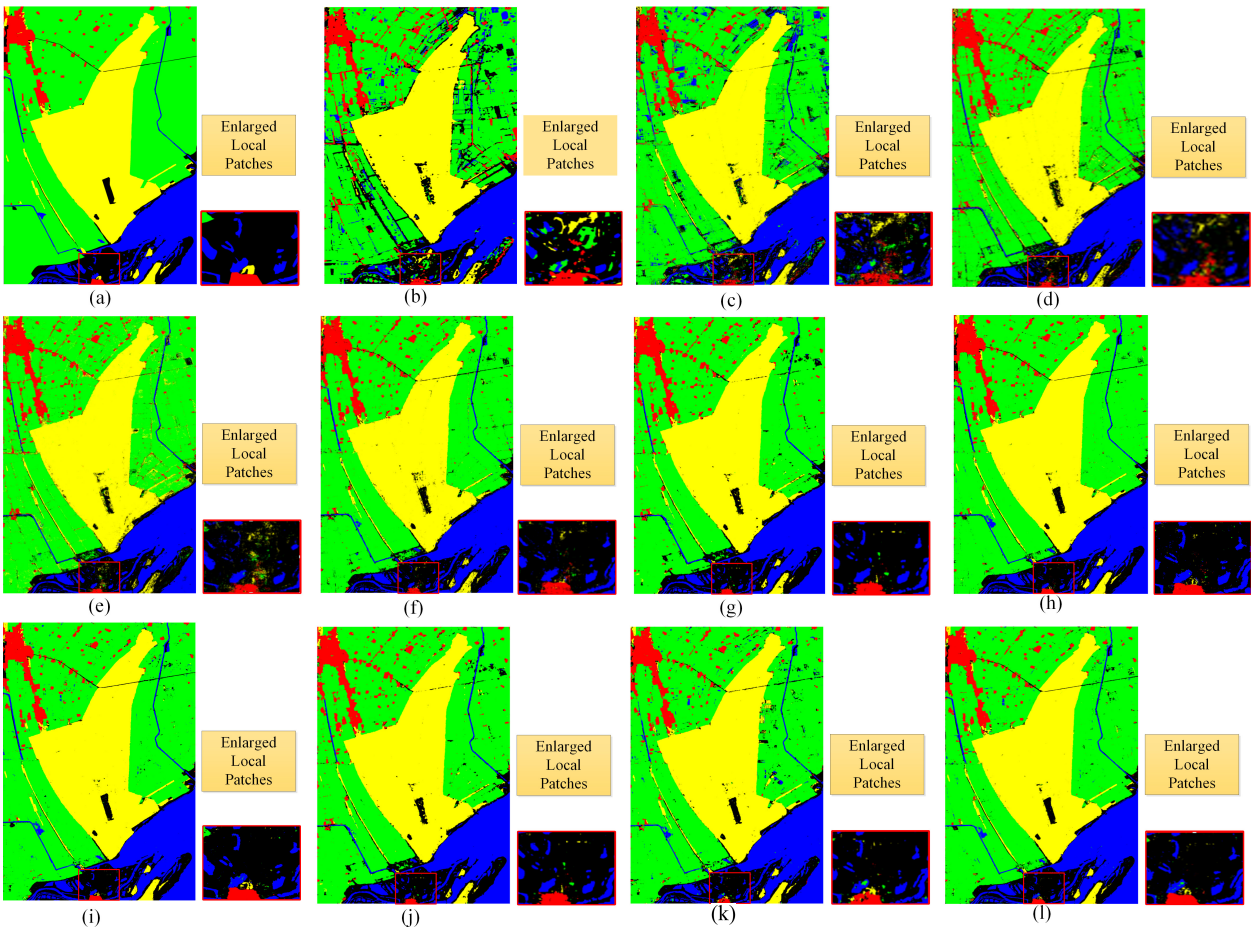

Fig. 9. Classification maps of different methods with missing SAR modality on PoDelta. (a) Ground truth. (b) SCNN [3]. (c) Suoyan*et.al* [66]. (d) 2-D-CNN [10]. (e) MDL-RS [45]. (f) MultiCNN [63]. (g) MBFNet [43] (ResNet34). (h) CHGFNet [46] (ResNet34). (i) Garcia *et. al* [30]. (j) Kampffmeyer *et.al* [38]. (k) DH-ADNet [52] (ResNet34). (l) DAGDNet (ResNet34).

from the teacher is used as auxiliary information, which only makes the student to obtain the performance over the reference methods, the SAR-branch or Opt.-branch2; the heterogeneous gap causes that using low-information SAR modality to imitate high information optical modality is difficult. Nevertheless, our DAGDNet relieves the heterogeneous gap by introducing a grouping distillation strategy, which obtains better performance than state-of-the-art multimodal knowledge distillation methods [30], [38], [52], i.e., row#21 versus row#17. Similarly, our proposed DAGDNet with missing optical modality obtains better performance than state-of-the-art multimodal knowledge distillation methods, i.e., row#19 versus row#13, and slightly lower than multimodal method, CHGFNet. The reasons are that we integrate the IG-FGM and the MS-ADL into the DAGDNet in an end-to-end manner, where the IG-FGM is responsible for training a good "teacher" to guide that the "student" learns privileged information, the MS-ADL completes adaptive feature grouping and multilayers feature distillation.

Visually, Figs. 7–9 show the classification score maps of the proposed DAGDNet and comparison methods on the Libourne, Bergerac, and PaDelta datasets, respectively, in which, the classification maps of knowledge distillation methods with missing SAR modality are present. Only using unimodal data might lead to weak classification maps with extensive misclassification. Also, our method can guide the "student" to learn privileged information, thereby reducing the misclassification and generating more smooth classification maps. To more intuitively compare classification maps of different methods, we focus on some detail with the red and blue regions in Figs. 7–9. Taking Fig. 7 as an example, in the red region, many of the resident area are not smooth, which were misclassified as the tree and farmland area by comparison methods, whereas the resident areas are accurately identified by the DAGDNet. In the blue regions, the proposed DAGDNet provides the smoothest visual results in comparison with other methods, and the farmland and tree areas are easily confused for the comparison methods. Similarly, we can also get a similar conclusion from Figs. 8 and 9, which demonstrates the effectiveness and superiority of our method in land cover classification with privileged modality.

D. Ablation Experiments

To ensure the contribution of the IG-FGM and the MS-ADL strategy for final land cover classification results, we further conduct ablation experiments, as shown in Table V. It records how our network successively improves the performance. For rows #1, #4, and #7, our DADGNet without the IG-FGM

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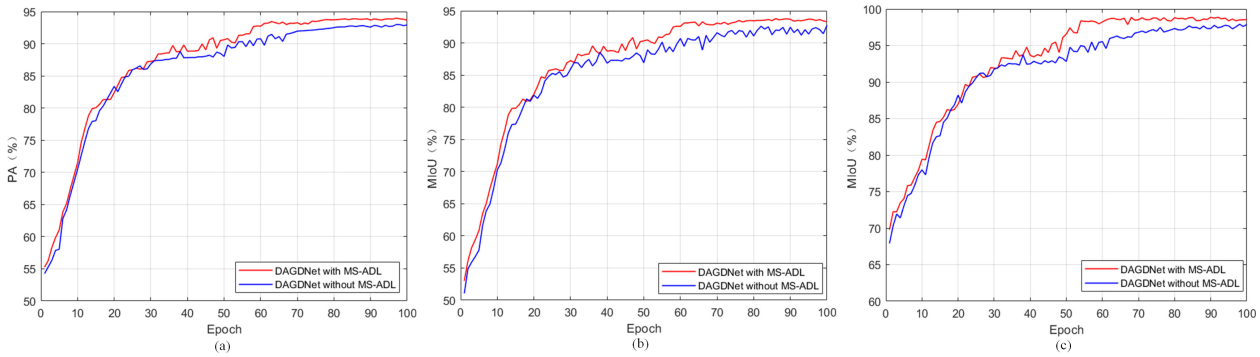

Fig. 10. Curves show the effectiveness of MS-ADL. The smoothed curves record the PA value with missing SAR modality during the training phase. (a) PA values on the Libourne dataset. (b) PA values on the Bergerac dataset. (c) PA values on the PoDelta dataset.

TABLE V

EFFECTIVENESS OF THE IG-FGM AND THE MS-ADL ON THE LIBOURNE, BERGERAC, AND PODELTA DATASETS

#	Datasets	IG-FGM	MS-ADL	Metrice(%)			
				PA	MPA	MIoU	FWIoU
1	Libourne			88.52	91.04	78.45	79.45
2		✓		90.45	93.98	81.01	81.95
3		✓	✓	**90.89**	**94.33**	**81.35**	**82.54**
4	Bergerac			88.45	92.67	88.09	80.01
5		✓		91.23	94.02	90.02	81.98
6		✓	✓	**91.52**	**94.56**	**90.33**	**82.36**
7	PoDelta			94.03	96.11	85.05	91.23
8		✓		96.01	96.56	87.09	93.23
9		✓	✓	**96.65**	**97.06**	**88.82**	**93.89**

and the MS-ADL is used as the baseline, and the objective function is obtained by removing the $\tilde{L}^{S}_{\mathrm{MSE}}$ item of (10). When the IG-FGM is introduced into the baseline in rows #2, #5, and #8, it achieves better results than the baseline since the IG-FGM progressively realizes the interaction between the optical and SAR features to guide the learning of the "student" network. Also, we further present the results of our DADGNet in rows #3, #6, and #9. It obtains significant improvements compared with its variants. Moreover, to more intuitively understand the effectiveness of the MS-ADL, we compare the effectiveness curves of the DADGNet with and without the MS-ADL strategy during the training phase, as shown in Fig. 10. It can be noticed that the curve of the DADGNet with the MS-ADL reaches a steady state faster than the DADGNet without the MS-ADL, which means that the MS-ADL strategy accelerates the network convergence. Therefore, it is seen that the MS-ADL strategy brings improvements.

In addition, we further study our DADGNet, which can promote the "student" network to learn privileged information under the IG-FGM and the MS-ADL strategy. Fig. 11 shows the histograms of different methods, where the SAR-branch and Opt.-branch denote the optical and SAR subnetworks from the "teacher," respectively, which belong to unimodal methods, and the "teacher" network belongs to multimodal methods. As expected, our DADGNet achieves significant improvement compared with the unimodal methods, and its results are slightly below the "teacher" network. It means that the IG-FGM and the MS-ADL are effective to learn privileged information.

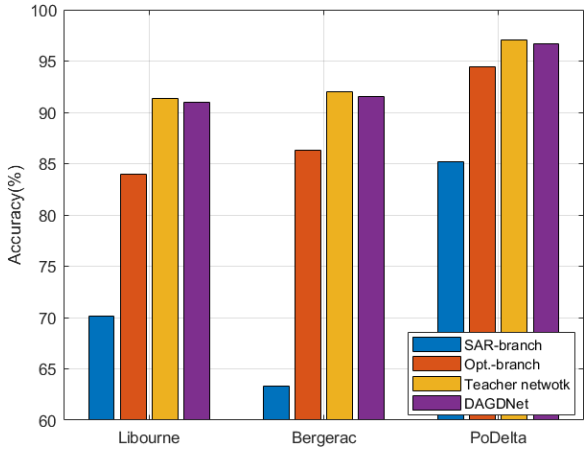

Fig. 11. Histograms show the effectiveness of the DAGDNet on the Libourne, Bergerac, and PoDelta datasets.

TABLE VI

PARAMETERS NUMBER OF DIFFERENT METHODS ON THE LIBOURNE, BERGERAC, AND PODELTA DATASETS

#	Methods	Backbones	Datasets		
			Libourne	Bergerac	PoDelta
			Model parameters (M)		
1	Garcia *et al.* [30]	AlexNet*	73.71	73.711	55.671
2	Kampffmeyer *et al.* [38]	AlexNet*	27.751	27.752	20.88
3	DH-ADNet [52]	AlexNet*	75.122	75.123	57.151
		ResNet34	171.504	171.505	171.503
4	DAGDNet (our)	AlexNet*	28.574	28.575	22.281
		ResNet34	69.155	69.157	69.154

E. Computational Complexity

Table VI presents the parameters number of the Garcia *et al.* [30], Kampffmeyer *et al.* [38], DH-ADNet [52], and the proposed DAGDNet on the Libourne, Bergerac, and PaDelta datasets. For a fair comparison, the DH-ADNet (ResNet34) uses the ResNet34 as the backbone of the optical, SAR, and hallucination branches, and the dynamic-hierarchical attention distillation module (DH-ADM) is embedded into the $layer1$–$layer4$. The IG-FGM is embedded into the $Conv1$, LRN_1, and LRN_2 of the AlexNet* in DAGDNet (AlexNet*). As shown in Table VI, first, the Garcia *et al.* [30] and DH-ADNet [52] methods considered the training paradigm from the perspective of curriculum

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learning, which have the most complex network structures, resulting in the parameters number of the Garcia *et al.* [30] and DH-ADNet [52] methods that have been greatly improved. Then, by comparing the Kampffmeyer *et al.* [38] and the proposed DAGDNet (AlexNet*), it is seen that the parameters number of the proposed DAGDNet (AlexNet*) has a slight increase since the IG-FGM is introduced. However, compared with the state-of-the-art privileged methods, the IG-FGM improves the performance of land cover classification with completely missing modalities, although it introduces an acceptable increase of modal parameters.

V. Conclusion

This article proposes a DAGDNet for land cover classification with privileged modality, which learns rich privileged information from all available modalities in train sets and improves the classification performance in test sets where one of the modalities data is scarce. The key of DAGDNet includes the IG-FGM and the MS-ADL. The IG-FGM decomposes multimodal features into modality-shared and modality-specific components to realize grouping distillation. Furthermore, to implement progressive blending of multimodalities, the IG-FGM is inserted into different layers of the "teacher" network. Also, the MS-ADL strategy adaptively highlights the importance of hierarchical features distillation and components distillation. Finally, we evaluate our approach on the MLCC task with privileged information and show that our model outperforms the state-of-the-art method on representative coregistered optical and SAR datasets.

References

[1] W. Zhao, Y. Qu, J. Chen, and Z. Yuan, "Deeply synergistic optical and SAR time series for crop dynamic monitoring," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111952.

[2] D. Lu and Q. Weng, "Use of impervious surface in urban land-use classification," *Remote Sens. Environ.*, vol. 102, nos. 1–2, pp. 146–160, May 2006.

[3] X. Liu, C. He, Q. Zhang, and M. Liao, "Statistical convolutional neural network for land-cover classification from SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 9, pp. 1548–1552, Sep. 2020.

[4] H. Zhang and R. Xu, "Exploring the optimal integration levels between SAR and optical data for better urban land cover mapping in the Pearl River Delta," *Interface J. Appl. Earth Obser. Geoinf.*, vol. 64, pp. 87–95, Feb. 2018.

[5] H. Guo, Q. Shi, B. Du, L. Zhang, D. Wang, and H. Ding, "Scene-driven multitask parallel attention network for building extraction in high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4287–4306, May 2021.

[6] H. Hishe *et al.*, "Analysis of land use land cover dynamics and driving factors in Desa'a forest in northern Ethiopia," *Land Use Policy*, vol. 101, Feb. 2021, Art. no. 105039.

[7] X. Wang, K. Tan, Q. Du, Y. Chen, and P. Du, "Caps-TripleGAN: GAN-assisted CapsNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7232–7245, Sep. 2019.

[8] N. Wambugu *et al.*, "Hyperspectral image classification on insufficient-sample and feature learning using deep neural networks: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102603.

[9] C. Sukawattanavijit, J. Chen, and H. Zhang, "GA-SVM algorithm for improving land-cover classification using SAR and optical remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 3, pp. 284–288, Mar. 2017.

[10] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 778–782, May 2017.

[11] B. Adriano *et al.*, "Learning from multimodal and multitemporal Earth observation data for building damage mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 132–143, May 2021.

[12] M. Agarwal, M. Leekha, R. Sawhney, and R. R. Shah, "Crisis-dias: Towards multimodal damage analysis-deployment, challenges and assessment," in *Proc. AAAI*, vol. 34, no. 1, Apr. 2020, pp. 346–353.

[13] V. Vapnik and A. Vashist, "A new learning paradigm: Learning using privileged information," *Neural Netw.*, vol. 22, nos. 5–6, pp. 544–557, 2009.

[14] P. J. García-Laencina, J.-L. Sancho-Gómez, and A. R. Figueiras-Vidal, "Pattern classification with missing data: A review," *Neural Comput. Appl.*, vol. 19, no. 2, pp. 263–282, 2010.

[15] J. G. Ibrahim, M.-H. Chen, S. R. Lipsitz, and A. H. Herring, "Missing-data methods for generalized linear models: A comparative review," *J. Amer. Stat. Assoc.*, vol. 100, no. 469, pp. 332–346, Mar. 2005.

[16] D. Ramachandram and G. W. Taylor, "Deep multimodal learning: A survey on recent advances and trends," *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 96–108, Nov. 2017.

[17] M. Das and S. K. Ghosh, "A deep-learning-based forecasting ensemble to predict missing data for remote sensing analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5228–5236, Dec. 2017.

[18] V. Sharmanska, N. Quadrianto, and C. H. Lampert, "Learning to transfer privileged information," 2014, *arXiv:1410.0389*.

[19] W. Xu, W. Liu, H. Chi, S. Qiu, and Y. Jin, "Self-paced learning with privileged information," *Neurocomputing*, vol. 362, pp. 147–155, Oct. 2019.

[20] H. Yang and I. Patras, "Privileged information-based conditional structured output regression forest for facial point detection," *IEEE Trans. Circu. Syst. Video Technol.*, vol. 25, no. 9, pp. 1507–1520, Sep. 2015.

[21] Z. Hui, X. Wang, and X. Gao, "Fast and accurate single image super-resolution via information distillation network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 723–731.

[22] J. Feyereisl, S. Kwak, J. Son, and B. Han, "Object localization based on structural SVM using privileged information," in *Proc. NeurIPS*, vol. 27, Dec. 2014, pp. 208–216.

[23] X. Yang, M. Wang, and D. Tao, "Person re-identification with metric learning using privileged information," *IEEE Trans. Image Process.*, vol. 27, no. 2, pp. 791–805, Feb. 2018.

[24] J. Feyereisl and U. Aickelin, "Privileged information for data clustering," *Inf. Sci.*, vol. 194, pp. 4–23, Jul. 2012.

[25] D. Lopez-Paz, L. Bottou, B. Scholkopfl, and V. Vapnik, "Unifying distillation and privileged information," 2015, *arXiv:1511.03643*.

[26] N. Passalis, M. Tzelepi, and A. Tefas, "Heterogeneous knowledge distillation using information flow modeling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2336–2345.

[27] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Interface J. Comput. Vis.*, vol. 129, no. 6, pp. 1789–1819, Mar. 2021.

[28] Y. Zhang, X. Li, and Z. Zhang, "Efficient person search via expert-guided knowledge distillation," *IEEE Trans. Cybern.*, vol. 51, no. 10, pp. 5093–5104, Oct. 2021.

[29] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[30] N. C. Garcia, P. Morerio, and V. Murino, "Modality distillation with multiple stream networks for action recognition," in *Proc. ECCV*, Sep. 2018, pp. 103–118.

[31] Y. Liu, K. Wang, G. Li, and L. Lin, "Semantics-aware adaptive knowledge distillation for sensor-to-vision action recognition," *IEEE Trans. Image Process.*, vol. 30, pp. 5573–5588, 2021.

[32] N. C. Garcia, P. Morerio, and V. Murino, "Learning with privileged information via adversarial discriminative modality distillation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 10, pp. 2581–2593, Jul. 2020.

[33] Y. Liu, K. Chen, C. Liu, Z. Qin, Z. Luo, and J. Wang, "Structured knowledge distillation for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2599–2608.

[34] Y. Wang, W. Zhou, T. Jiang, X. Bai, and Y. Xu, "Intra-class feature variation distillation for semantic segmentation," in *Proc. ECCV*, Cham, Switzerland: Springer, Aug. 2020, pp. 346–362.

[35] Z. Luo, J. T. Hsieh, L. Jiang, J. C. Niebles, and L. Fei-Fei, "Graph distillation for action detection with privileged modalities," in *Proc. ECCV*, Aug. 2018, pp. 166–183.

[36] S. Chen and Q. Zhao, "Attention-based autism spectrum disorder screening with privileged modality," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1181–1190.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:18:30 UTC from IEEE Xplore. Restrictions apply.

[37] S. Kumar, B. Banerjee, and S. Chaudhuri, "Improved landcover classification using online spectral data hallucination," *Neurocomputing*, vol. 439, pp. 316–326, Jun. 2021.
[38] M. Kampffmeyer, A.-B. Salberg, and R. Jenssen, "Urban land cover classification with missing data modalities using deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 1758–1768, Jun. 2018.
[39] S. Pande, A. Banerjee, S. Kumar, B. Banerjee, and S. Chaudhuri, "An adversarial approach to discriminative modality distillation for remote sensing image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 4571–4580.
[40] J. Hoffman, S. Gupta, and T. Darrell, "Learning with side information through modality hallucination," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 826–834.
[41] P. V. N. Rao *et al.*, "Textural analysis of IRS-1D panchromatic data for land cover classification," *Int. J. Remote Sens.*, vol. 23, no. 17, pp. 3327–3345, Jan. 2002.
[42] R. Fernandez-Beltran, J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and F. Pla, "Remote sensing image fusion using hierarchical multimodal probabilistic latent semantic analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4982–4993, Dec. 2018.
[43] X. Li, L. Lei, Y. Sun, M. Li, and G. Kuang, "Multimodal bilinear fusion network with second-order attention-based channel selection for land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1011–1026, 2020.
[44] X. Liu, L. Li, F. Liu, B. Hou, S. Yang, and L. Jiao, "GAFNet: Group attention fusion network for PAN and MS image high-resolution classification," *IEEE Trans. Cybern.*, early access, Mar. 30, 2021, doi: 10.1109/TCYB.2021.3064571.
[45] D. Hong *et al.*, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.
[46] X. Li, L. Lei, Y. Sun, M. Li, and G. Kuang, "Collaborative attention-based heterogeneous gated fusion network for land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 3829–3845, May 2021.
[47] D. Hong, J. Hu, J. Yao, J. Chanussot, and X. X. Zhu, "Multimodal remote sensing benchmark datasets for land cover classification with a shared and specific feature learning model," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 68–80, Aug. 2021.
[48] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.
[49] R. Hänsch and O. Hellwich, "Fusion of multispectral LiDAR, Hyperspectral, and RGB data for urban land cover classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 2, pp. 366–370, Feb. 2021.
[50] N. Audebert, B. L. Saux, and S. Lefèvre, "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 140, pp. 20–32, Jun. 2018.
[51] M. J. Steinhausen, P. D. Wagner, B. Narasimhan, and B. Waske, "Combining Sentinel-1 and Sentinel-2 data for improved land use and land cover mapping of monsoon regions," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 73, pp. 595–604, Dec. 2018.
[52] X. Li, L. Lei, Y. Sun, and G. Kuang, "Dynamic-hierarchical attention distillation with synergetic instance selection for land cover classification using missing heterogeneity images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, doi: 10.1109/TGRS.2020.3048332.
[53] W. Lee, J. Lee, D. Kim, and B. Ham, "Learning with privileged information for efficient image super-resolution," in *Proc. ECCV*, Aug. 2020, pp. 465–482.
[54] L. Bottou, "Stochastic gradient learning in neural networks," *Proc. Neuro-Nimes*, vol. 91, no. 8, p. 12, 1991.
[55] Y. Yao, F. Shen, J. Zhang, L. Liu, Z. Tang, and L. Shao, "Extracting privileged information for enhancing classifier learning," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 436–450, Jan. 2019.
[56] H. Hoefling, "A path algorithm for the fused lasso signal approximator," *J. Comput. Graph. Statist.*, vol. 19, no. 4, pp. 984–1006, 2010.
[57] J. T. F. Hastie, H. Höfling, and R. Tibshirani, "Pathwise coordinate optimization," *Ann. Appl. Statist.*, vol. 1, no. 2, pp. 302–332, 2007.
[58] L. Han and Y. Zhang, "Discriminative feature grouping," in *Proc. AAAI*, vol. 29, no. 1, Feb. 2015, pp. 1–7.
[59] T. Zhang, G. Qi, B. Xiao, and J. Wang, "Interleaved group convolutions," in *Proc. ICCV*, Venice, Italy, Oct. 2017, pp. 4383–4392.
[60] X. Zhao, L. Sang, G. Ding, Y. Guo, and X. Jin, "Grouping attribute recognition for pedestrian with joint recurrent learning," in *Proc. IJCAI*, Jul. 2018, pp. 3177–3183.
[61] Y. Y. Zhang, H. Wang, X. Lv, and P. Zhang, "Capturing the grouping and compactness of high-level semantic feature for saliency detection," *Neural Netw.*, vol. 142, pp. 351–362, Oct. 2021.
[62] T. Kobayashi, "Group softmax loss with discriminative feature grouping," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 2614–2623.
[63] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[64] OpenStreetMap. (2017). *Planet Dump Retrieved From*. [Online]. Available: https://www.openstreetmap.org
[65] W. Jin, M. Sanjabi, S. Nie, L. Tan, X. Ren, and H. Firooz, "Modality-specific distillation," 2021, *arXiv:2101.01881*.
[66] S. Pan *et al.*, "Land-cover classification of multispectral LiDAR data using CNN with optimized hyper-parameters," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 241–254, Aug. 2020.

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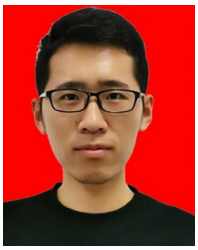

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